Operating Systems Lab Assignment

Experiment: Hive Containment Protocol

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Platform: Ubuntu Linux

Part A: Execution and Observation

Screenshot 1: Hive Startup

```
vboxuser@Ubuntu-24:~/Desktop$ cd os-lab
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ls
hive.c
vboxuser@Ubuntu-24:~/Desktop/os-lab$ gcc hive.c -o hive
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ./hive PES2UG23CS277
[Hive 5057] Starting. SRN=277, L=5, Z=6, M=67
[Hive 5057] Home Directory: /home/vboxuser
[Hive 5057] Current Directory: /home/vboxuser/Desktop/os-lab
[Hive 5057] Spawned 5 larvae and 6 husks.
[Hive 5057] Control commands: type 'reap' to reap zombies, 'quit' to shutdown.
[Hive 5057] >
```

Screenshot 2: Reading Hive PID File

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ cat /tmp/lab_277/hive.pid
5246
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 3: Process Tree Showing Larvae and Husks

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ps -eo pid,ppid,stat,cmd | grep 5246
   5246    5234 S+   ./hive PES2UG23CS277
   5247    5246 S+   ./hive PES2UG23CS277
   5248    5246 S+   ./hive PES2UG23CS277
   5249    5246 S+   ./hive PES2UG23CS277
   5250    5246 S+   ./hive PES2UG23CS277
   5251    5246 S+   ./hive PES2UG23CS277
   5252    5246 Z+   [hive] <defunct>
   5253    5246 Z+   [hive] <defunct>
   5254    5246 Z+   [hive] <defunct>
   5255    5246 Z+   [hive] <defunct>
   5256    5246 Z+   [hive] <defunct>
   5257    5246 Z+   [hive] <defunct>
   5304    5261 S+   grep --color=auto 5246
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 4: Zombie Processes (Husks)

```
[Hive 5246] > reap
[Hive 5246] Command received: reap (reaping zombies)...
[Hive 5246] Reaped 6 zombie(s).
[Hive 5246] >
```

Screenshot 5: Memory Usage of Larvae

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ps -eo stat,cmd | grep Z
S+   grep --color=auto Z
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 6: Larva Log File Contents

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ps -eo pid,ppid,stat,cmd | grep larva
   5373    5261 S+   grep --color=auto larva
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 7: Reaping Zombies

```
[Hive 5246] > quit
[Hive 5246] Command received: quit (shutting down)...
[Hive 5246] Shutdown complete.
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 8: Verification – No Zombies

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ps -eo pid,ppid,stat,cmd | grep 5246
   5770    5261 S+   grep --color=auto 5246
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 9: Larvae Still Running

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ps -eo pid,ppid,stat,cmd | grep 5831
   5831    5234 S+   ./hive PES2UG23CS277
   5832    5831 S+   ./hive PES2UG23CS277
   5833    5831 S+   ./hive PES2UG23CS277
   5834    5831 S+   ./hive PES2UG23CS277
   5835    5831 S+   ./hive PES2UG23CS277
   5836    5831 S+   ./hive PES2UG23CS277
   5837    5831 Z+   [hive] <defunct>
   5838    5831 Z+   [hive] <defunct>
   5839    5831 Z+   [hive] <defunct>
   5840    5831 Z+   [hive] <defunct>
   5841    5831 Z+   [hive] <defunct>
   5842    5831 Z+   [hive] <defunct>
   5862    5261 S+   grep --color=auto 5831
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Screenshot 10: Graceful Shutdown of Hive

```
vboxuser@Ubuntu-24:~/Desktop/os-lab$ ./hive PES2UG23CS277
[Hive 5831] Starting. SRN=277, L=5, Z=6, M=67
[Hive 5831] Home Directory: /home/vboxuser
[Hive 5831] Current Directory: /home/vboxuser/Desktop/os-lab
[Hive 5831] Spawned 5 larvae and 6 husks.
[Hive 5831] Control commands: type 'reap' to reap zombies, 'quit' to shutdown.
[Hive 5831] > quit
[Hive 5831] Command received: quit (shutting down)...
[Hive 5831] Shutdown complete.
vboxuser@Ubuntu-24:~/Desktop/os-lab$
vboxuser@Ubuntu-24:~/Desktop/os-lab$
```

Part B: Theory Questions and Answers

Q: How can you verify the parent/child relationship between the Hive and its children?

A: Each process in Linux has a PID and a PPID. PID uniquely identifies a process, while PPID identifies its parent. By using commands like ps -eo pid,ppid,cmd or pstree, we can observe that all larvae and husk processes have their PPID equal to the Hive PID, confirming the parent-child relationship.

Q: Why do zombie processes exist until the parent reaps them?

A: When a child process exits, the kernel retains its exit status so that the parent can read it. Until the parent calls wait() or waitpid(), the child remains as a zombie occupying a process table entry.

Q: What is the difference between a zombie process and a running process?

A: A zombie process has completed execution and does not consume CPU or memory, but still occupies an entry in the process table. A running process is actively executing and consumes system resources.

Q: Why might a parent use a non-blocking wait (WNOHANG)?

A: A non-blocking wait allows the parent to continue execution without getting stuck waiting for child processes to terminate. This is useful when the parent must remain responsive.

Q: Explain the lifecycle of the husk processes.

A: Husk processes are created using fork() and immediately exit. Since the parent does not reap them immediately, they become zombie processes until the reap command is issued.

Q: Why do larvae enter pause() instead of exiting after memory allocation fails?

A: By entering pause(), larvae remain alive without consuming CPU resources while still occupying memory. If larvae exited, the Hive would lose its intended memory pressure behavior.